

H1 / H2 / H3 Matrix Report

Source: results.csv · cells: 40 · targets: 10 · conditions: full, no-fusion, no-replan, random-order

Executive Summary

Total cells	40
Distinct targets	10
Mean duration / cell	89.1s
Total findings across cells	7126
Engine pool example	nmap,url_crawler,http_probe,nikto,header_vuln_check

Findings per (target × condition)

Target	full	no-fusion	no-replan	random-order
bodgeit	29.0	28.0	14.0	29.0
bwapp	33.0	33.0	14.0	33.0
commix_testbed	39.0	39.0	14.0	39.0
dvwa_live	2033.0	2033.0	14.0	2033.0
juiceshop	44.0	44.0	14.0	44.0
nowasp	40.0	40.0	14.0	40.0
pygoat	25.0	23.0	14.0	30.0
skf_csrf	26.0	26.0	14.0	26.0
vampi	31.0	31.0	14.0	31.0
webgoat	28.0	28.0	14.0	28.0

H1 — Closed-loop replanning improves coverage

mean(full)	232.8
mean(no-replan)	14.0
paired t (all)	t=1.094, p=0.1512
Wilcoxon (all)	W=55.0, p=0.000977
paired t (no DVWA)	t=8.357, p=0.000016, Cohen d=2.786
mean lift (no DVWA)	+134%

H2 — Cross-tool fusion

mean(full)	232.8
mean(no-fusion)	232.5
paired t	t=1.406, p=0.0967
Wilcoxon	W=3.0, p=0.250000

H3 — Decision-quality stratification

Condition	Correct	Suboptimal	Incorrect	Total	% incorrect
full	50	0	20	70	28.6%
no-fusion	50	0	20	70	28.6%
no-replan	30	0	0	30	0.0%
random-order	50	0	10	60	16.7%